CHONG HAN
to me, Oyakhi, Tolulope

12:47 AM (21 hours ago)

Hello James,

Quick clarification on where my cointegration statements came from:

Short note: My initial crisis-specific comments summarized the paper's existing results and Prof. Ibhagui's guidance, with plots used only for intuition—I did not run new tests at that time. Per your follow-up, I have now extracted cointegration information from the FX log-levels using our Methodology (unit root tests to confirm I(1), then system tests for long-run relations). Under the current Python settings, the rank comes out 0 across episodes; I'll align the exact window/lag/deterministic terms with Prof.'s specification to match the reported GFC rank.

I extract cointegration as in our Methodology Section 6:

- Work with the log spot rates from the Excel sheets (EUR/USD, JPY/USD, GBP/USD, CHF/USD, AUD/USD), so five columns in total.
- Run unit root tests to confirm the series are I(1), I got all the P values are >0.05
- On each crisis subperiod, run a system cointegration test on the log-levels to identify the cointegration rank

Rank status:

- Dot-Com (2000–2002):
  – Current replication: rank = 0 (no long-run cointegration).
  – Consistency: matches my earlier "lack of strong long-run cointegration." No discrepancy here.
- GFC (2007–2009):
  – Current replication (default Python spec): rank = 0.
  – Prior statement/reference: Prof. Ibhagui's EViews reports rank = 1.
  – Likely reason for the mismatch: Johansen is sensitive to exact sample window, lag length, and deterministic component (e.g., restricted constant vs. constant/trend). **Please let me know your setting if you derive better results, I've tried multiple times and keep getting 0, I doubt I had set differently.**
- Covid (2020):
  – Current replication: rank = 0.
  – Consistency: matches my earlier "no cointegration." No discrepancy here.

Best,
Chong

James Evans
to CHONG, Oyakhi, Tolulope

10:36 PM (0 minutes ago)

Hello Chong,

Thank you for sharing your analysis; it is very supportive of Proposition 2!

Lower cointegration during crises is consistent with Proposition 2. If all currencies load heavily on the same global factors in nearly the same way, then their differences (linear combinations) do not cancel out. This means fewer stationary relations and therefore a lower cointegration rank. Proposition 2 predicts exactly this outcome: crises compress autonomy, so currencies are increasingly driven by the same global shocks. In this environment, it is not possible to form many independent stationary spreads across currencies, and the data reflect this as a low cointegration rank.

Great Stuff!

Best,

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Reply Reply all Forward